Wintertime Antarctic coastal polynya area: 1992–2008

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[1] An iterative classification technique based on Special Sensor Microwave/Imager data is used to estimate the total daily circum-Antarctic coastal polynya area for 1992–2008. The average wintertime (June to September) area is estimated at 245,000 ± 10,000 km^2, and varies between 210,000 km^2 (1997) and 281,000 km^2 (2007). The polynyas along East-Antarctica (60 to 160°E) contribute about 40% to this area. The most persistent polynyas are located along East Antarctica: Lars-Christensen Coast (LCC), Prydz Bay, Western Davis Sea, Mertz Glacier, and in the Ross Sea: Ross Ice Shelf and Terra Nova Bay. The polynya at the LCC is observed on 110 ± 5 days during winter for 1992–2008. It is the most persistent of the observed polynyas, covering an average area of 2400 km^2 on more than 90 days. Trends in the average wintertime area of the major coastal polynyas during 1992–2008 are small and not significant. **Citation:** Kern, S. (2009), Wintertime Antarctic coastal polynya area: 1992–2008, *Geophys. Res. Lett.*, *36*, L14501, doi:10.1029/2009GL038062.

1. Introduction

[2] Coastal polynyas are ubiquitous around Antarctica during winter [*Morales Maqueda et al.*, 2004; *Massom et al.*, 1998]. They are often caused by offshore blowing katabatic winds that push the sea ice away from the coast or shelf ice border. Latent heat released during ice formation in these polynyas helps to keeping them open so that they are called latent heat polynya in the absence of a considerable oceanic heat flux.

[3] Water mass modification caused by ice formation in such polynyas during winter, e.g., the Terra Nova Bay polynya, and the Mertz Glacier polynya, is known to contribute to Antarctic Bottom Water (AABW) formation [*Marsland et al.*, 2004; *Van Woert*, 1999]. During winter the heat and moisture input into the atmosphere over polynyas is larger compared to the surrounding thicker ice. During spring enhanced absorption of solar energy triggers the melt-onset. A better knowledge of the spatio-temporal variability of the Antarctic coastal polynya area distribution could help to better understand the involved ocean-ice-atmosphere interactions and to provide a quantitative prerequisite to estimate the associated ice production.

[4] Typically, Antarctic coastal polynyas are small-scale features, which are difficult to monitor from space. One approach to estimate the polynya area is to take sea-ice concentration maps obtained from (coarse-resolution) satellite microwave radiometry and to apply an ice-concentration threshold to determine the open water fraction [*Parmiggiani*, 2006; *Massom et al.*, 1998].

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[5] Particularly new and thin ice types, which are common for a polynya, are often under-represented in standard ice-concentration retrieval algorithms [*Shokr et al.*, 2009; *Kwok et al.*, 2007]. As an alternative the Polynya Signature Simulation Method (PSSM) [*Markus and Burns*, 1995; *Hunewinkel et al.*, 1998] could be applied. This method classifies observed brightness temperatures into surface types rather than calculating an ice concentration. The PSSM has already been used in a number of previous studies [*Kern et al.*, 2005; *Arrigo and van Dijken*, 2003; *Renfrew et al.*, 2002].

[6] The goal of this paper is to present a temporally and spatially consistent time-series of the circum-Antarctic polynya area distribution during wintertime (June to September). In order to achieve this goal, the PSSM is applied to Special Sensor Microwave/Imager (SSM/I) data to obtaining maps of the polynya distribution around Antarctica for the period 1992–2008. Swath data are used instead of daily averages in order to account for sub-daily variations in the polynya area caused by, for example, rapidly changing surface winds, or tides. These maps reveal the typical polynya location and distribution, and allow us to draw conclusions about the variability of the polynya area and the persistence of the major polynyas within the considered 17-year period.

2. Method

[7] The PSSM uses an iterative classification technique applied to the SSM/I brightness temperature polarization difference measured at frequencies of 37 GHz, $P(37)$, and 85.5 GHz, $P(85)$, and normalized with the brightness temperature polarization sum at the respective frequency to obtain the distribution of open water, thin and thick ice. Single-swath SSM/I brightness temperatures are interpolated onto a polar stereographic grid [*National Snow and Ice Data Center* (*NSIDC*), 1996] with 12.5 km × 12.5 km (37 GHz data) and 5 km × 5 km (85.5 GHz data) grid cell size using the Backus-Gilbert interpolation technique [*Stogryn*, 1978]. The brightness temperatures are corrected for intersensor differences because data acquired by SSM/I sensors aboard different Defense Meteorological Satellite Program spacecraft (F10, F13, and F15) are used [*Kern et al.*, 2007].

[8] A set of $n = 8$ binary maps with 5 km × 5 km grid cell size is obtained by separating thick ice and thin ice/open water in the $P(85)$ map using a set of n $P(85)$ threshold values. These are calculated as the differences between a (theoretical) maximum $P(85)$ value ($P(85)_{max}$ = 0.2) and $n - 1$ times the difference between $P(85)_{max}$ and a (theoretical) minimum $P(85)$ value ($P(85)_{min}$ = 0.01), divided by n; n is the number of equidistant steps for which a classification will be carried out. Each map is used to generate a synthetic map of $P(37)$ by assigning typical values of $P(37)$ (see below) to the two used surface

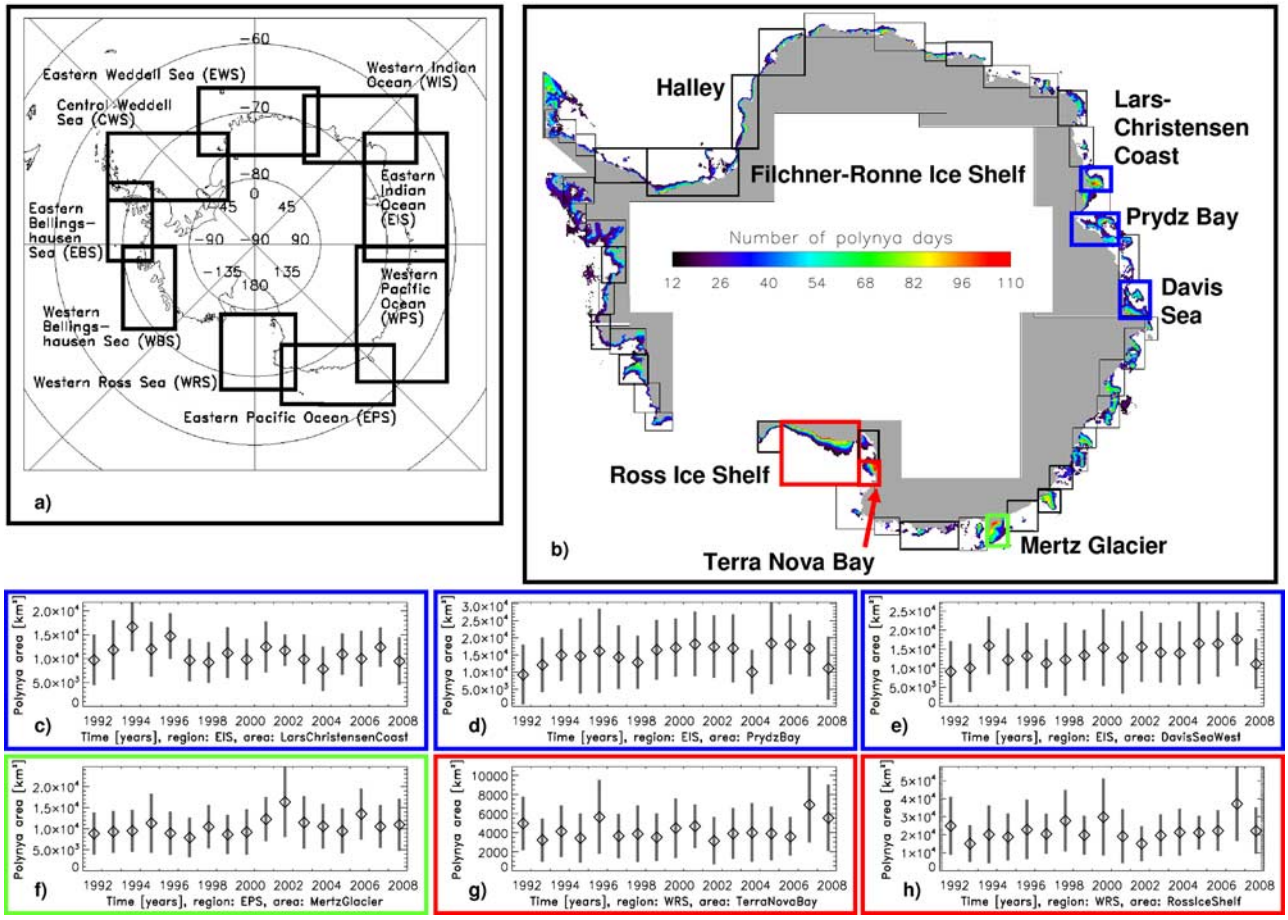

Figure 1. (a) Map of Antarctica with the regions of interest. (b) Wintertime circum-Antarctic polynya distribution for 2008. Shown is the number of days during which thin ice and/or open water is identified for each grid cell within the period June 1 to September 30 (polynya days). White (grey) areas denote missing data (land). Grid cells with less than 12 polynya days are also shown white. Black boxes denote sub-regions used to calculate the total polynya area. (c–h) Average wintertime polynya area of selected sub-regions (colored boxes in Figure 1b) for 1992–2008. For each year the average polynya area (diamonds) and its standard deviation (vertical grey bars) is given. Note the different y-axis scaling.

type classes: thick ice and thin ice/open water. Each synthetic map is convoluted with the SSM/I antenna pattern at 37 GHz, and is compared to the measured $P(37)$ by means of the root-mean square error (RMSE) and a linear correlation. The classification is repeated by taking the $P(85)$ threshold which resulted in the highest correlation (smallest RMSE) during the last iteration plus/minus the last stepsize as new values for $P(85)_{max}$ and $P(85)_{min}$, re-calculating the stepsize and n new $P(85)$ threshold values, and doing the classification until the RMSE (correlation) has converged to a constant minimum (maximum) value. The method is described in detail by *Markus and Burns* [1995]. The two-class distribution given by the synthetic $P(37)$ map is assumed to represent the wanted surface type distribution [see *Markus and Burns*, 1995; *Hunewinkel et al.*, 1998].

[9] Land/Ice shelf grid cells and grid cells outside the swath are not included. The typical value used for the synthetic distribution of $P(37)$ for thin ice/open water is 0.08. The value of $P(37)$ for thick ice is variable because it is calculated daily from the area with at least 90 % sea ice concentration within each ROI using the Comiso-Bootstrap algorithm [*Comiso et al.*, 1997]. The aim of using daily varying typical values for $P(37)$ of thick ice is to reduce the influence of weather-induced changes in the observed $P(37)$-values on the classification result. If, for example, a fixed value of 0.04 is used and the actual value of $P(37)$ exceeds 0.04 because of weather influences, then the thin ice area would be substantially overestimated [*Kern et al.*, 2007].

[10] An optimized land mask is used, in order to optimize the polynya distribution retrieval near the coast. This mask is generated by manually correcting the land-mask provided by *NSIDC* [1996] using clear-sky images acquired with channels 1 and 4 of the Advanced Very High Resolution Radiometer (AVHRR) interpolated onto a polar-stereographic grid with 1 km × 1 km grid-cell size. An optimized mask is generated for 2002 for all regions (Figure 1a) and is updated every other year for 1992–2004 for regions with a dynamic shelf ice border (regions CWS, EIS, EPS, and WRS) [*Kern et al.*, 2007].

[11] Ice bergs and occasionally also land-fast sea ice can exhibit values of $P(85)$ which are similar to those of thin ice. This causes mis-classifications of these surface types as thin ice. A filter is developed based on the observation that compared to a polynya P-values of fast ice and ice bergs

Table 1. Average Wintertime Polynya Area A and Its Standard Deviation σ_A for 1992–2008 for Each Region Given in Figure 1a Using Only Grid Cells of the Sub-regions Marked by the Boxes in Figure 1b

Region	CWS	EWS	WIS	EIS	WPS	EPS	WRS	WBS	EBS	Total
A [10^3 km^2]	21.2	22.0	17.8	51.8	33.1	26.8	35.2	21.0	16.2	245.2
σ_A [10^3 km^2]	3.8	3.1	2.8	4.3	2.8	2.8	4.1	2.8	4.0	10.2

vary less with time. For each grid cell the mean P-value of the two-weeks period preceding the day of interest and the associated temporal variability (variance) is calculated. By using a threshold value for the variance a mask is generated containing those grid cells which exhibit relatively high but temporally stable P-values. This mask (called P2week-mask henceforth) is subsequently used to flag grid cells potentially covered by ice bergs and/or fast ice in the polynya distribution map [see *Kern et al.*, 2007]. The P2week-mask fails to work properly if P-values observed over a polynya remain too stable. Such cases are identified manually in movies of the polynya distribution maps. The P2week-mask is modified accordingly to represent a longer time period (e.g., three weeks instead of two weeks).

[12] The obtained sub-daily polynya area has been verified with independent data. Regardless of the size of an individual polynya a value of 250 km^2 (ten polynya map grid cells) has to be considered as the mean error for the present study, i.e., taking into account the entire Antarctic. This value is larger than the one reported by *Markus and Burns* [1995]: 80 km^2, and results from the larger uncertainties arising from weather influences and the tendency of the radiometric signal to be similar for ice bergs, fast ice and thin ice. The minimum observable day-to-day change in the area of a polynya is found to be of the order of 100 km^2, which agrees with the value given by *Markus and Burns* [1995].

[13] The polynya area distribution is obtained separately for each ROI for the period April 1 to September 30 for the years 1992–2008. In the present paper, only the results of June to September are considered, because the time of freeze-up varies along the Antarctic coast and thus does the polynya formation, which starts rather late in regions WIS and EBS (Figure 1a).

3. Results

[14] Wintertime (wintertime is used instead of June–September henceforth) polynya distribution maps are obtained by counting the days during which a polynya is identified in the grid cells of each sub-region during wintertime (polynya days). Figure 1b shows such a wintertime polynya distribution map for 2008. All major coastal polynyas reported in previous studies [e.g., *Tamura et al.*, 2008; *Massom et al.*, 1998] are captured well. Polynyas that are situated in front of an ice shelf and are thus elongated, e.g., the Ross Ice Shelf polynya, can be discriminated from polynyas which develop in the lee of an ice tongue or fast-ice massif, e.g. the Terra Nova Bay, and Mertz Glacier polynyas. Maps of the kind shown in Figure 1b are produced for every winter of the period 1992–2008.

[15] Subsequently the mean wintertime polynya area is calculated for each sub-region indicated by the boxes in Figure 1b. It is given in Table 1. The two easternmost sub-regions of region EBS: Rothera and Cape Byrd, are not considered in this calculation, because the ice cover variations in these two sub-regions are dominated by variations in the ice edge position too often.

[16] The total circum-Antarctic mean wintertime polynya area is estimated at about 245 000 km^2. The regions bordering East-Antarctica (EIS, WPS, and EPS) contribute about 40% to this area. The regions feeding the Ross and Weddell Seas (WRS, CWS, and EWS) contribute together about 30%. Examples for the interannual variability of the wintertime polynya area is given in Figures 1c–1h for selected sub-regions (colored boxes in Figure 1b). Albeit a few sub-regions seem to reveal a gradual change in the wintertime polynya area (Figures 1c–1e) no statistical significant trend could be delineated.

[17] Using maps of the kind shown in Figure 1b the maximum number of days during which a polynya is observed is identified separately for each sub-region. Subsequently, for each sub-region the 17 (1992–2008) values of the maximum number of polynya days are averaged to obtain estimates of the average maximum number of polynya days for the period 1992–2008. The polynyas in sub-regions Lars-Christensen Coast, Prydz Bay, Davis Sea (region EIS), Mertz Glacier (region EPS), and Terra Nova Bay and Ross Ice Shelf (region WRS) have the highest persistence with average maximum polynya day counts above 100 (Figure 2a, red bars). This persistence is underlined by a standard deviation below ten days. Note that these polynyas also have the largest average wintertime polynya area (given by the width of the bars) compared to the other sub-regions, except the Terra Nova Bay polynya. Only slightly less persistent polynyas (at least 90 polynya days) can be identified in regions EWS and WPS (Figure 2a, orange bars).

[18] The average area that is covered during at least 90 days every winter during 1992–2008 by the most persistent polynyas (those six ones with more than 100 polynya days) varies between about 600 km^2 (Davis Sea) and 2400 km^2 (Lars-Christensen Coast) and shows a high inter-annual variability (Figure 2b). For sub-regions Lars-Christensen Coast, Terra Nova Bay, and Mertz Glacier, this area corresponds to 22%, 18%, and 16% of the average wintertime polynya area, respectively.

4. Discussion and Conclusion

[19] The Polynya Signature Simulation Method (PSSM) is applied to swath data of the SSM/I for 1992–2008 to obtain a time-series of the wintertime (June–Sep.) daily Antarctic polynya distribution. Based on this time-series, the total average wintertime circum-Antarctic polynya area is calculated: $(245 \pm 10) \times 10^3$ km^2. This corresponds to about 1.5% of the mean 1979–2006 total Antarctic sea ice area [*Cavalieri and Parkinson*, 2008]. East Antarctic coastal polynyas contribute about 40% to the total area. The Weddell and Ross Seas together contribute about 30%.

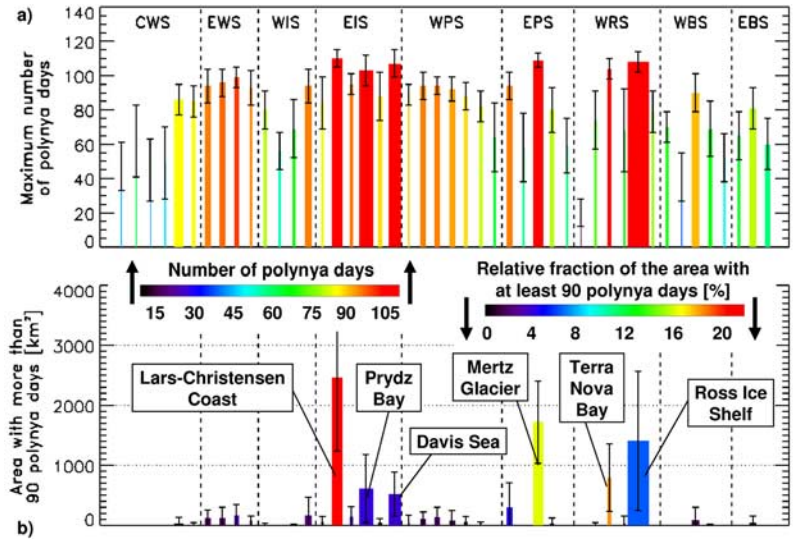

Figure 2. (a) Average maximum number of polynya days for 1992–2008. (b) Average area with at least 90 polynya days for 1992–2008. The width of all bars is scaled with the average wintertime polynya area of the respective sub-region, which is largest for the Ross Ice Shelf: $22.3 \pm 5.4 \times 10^3$ km^2. Colors in Figure 2a denote the number of polynya days (left color legend). Colors in Figure 2b give the fraction of the area with 90 polynya days relative to the average wintertime polynya area of that sub-region (right color legend). Black error bars at the end of each bar denote one standard deviation of the shown quantity.

[20] Some of the polynyas are investigated for the first time in this study so that a comparison with previous studies is not possible. However, for a few selected polynyas a number of modeling and observational studies exist. For the Mertz Glacier polynya *Marsland et al.* [2004] reported an average wintertime polynya area of 10 000 – 13 000 km^2 for 1991–2000 as a result from numerical modeling. This range compares reasonably well with the results of the present paper for 1992–2001: 9 700 ± 1300 km^2.

[21] Table 2 shows a comparison between the average wintertime polynya area derived with the PSSM for sub-regions Ross Ice Shelf, McMurdo Sound, and Terra Nova Bay (see Figures 1b, 1g, and 1h) and the corresponding values given by *Martin et al.* [2007] for 1992–2002. They defined the polynya area by applying a threshold to the ice-thickness distribution obtained from 37 GHz SSM/I data. Values for the McMurdo Sound polynya agree perfectly well. Values for the Ross Ice Shelf polynya found in this study are a bit smaller but agree with the results of *Martin et al.* [2007] within one standard deviation. The observed discrepancy could be caused by the shorter period considered in the present study which does not take April and May into account, and therefore misses the commonly larger polynya area during the beginning of winter. The Terra Nova Bay polynya area is overestimated by this study compared to *Martin et al.* [2007] (Table 2). This can be explained by i) the smaller size of the sub-region used by *Martin et al.* [2007], and ii) the more accurate land mask used in the present study. According to *Martin et al.* [2007] their method tends to underestimate the real polynya area by up to 30% in case of a small polynya.

[22] According to *Martin et al.* [2007] (*Tamura et al.* [2008]) the annual Ross Ice Shelf polynya area has increased (decreased) during the period 1992–2002 (1992–2001) with different trends in the associated ice production and different explanations for the observed AABW freshening [*Rintoul*, 2007]. In the present paper the wintertime Ross Ice Shelf polynya area shows no significant trend for 1992–2008 (Figure 2h). If considering a shorter period, like 1992–2001/02, then the observed wintertime polynya area trends become larger but not significant (Figure 2c) to h). However, the sign of the polynya-area trends observed in the sub-regions shown in Figures 1c–1h matches that of the ice-production trends found by *Tamura et al.* [2008]. The annual average polynya area is directly linked to the annual cumulative ice production, at least for the Ross Ice Shelf polynya [*Martin et al.*, 2007]. Therefore the observed lack of a significant trend in the wintertime polynya area during 1992–2008 could be interpreted as a missing significant change in the Antarctic coastal polynya ice production.

[23] The most productive polynyas identified by *Tamura et al.* [2008] reveal a large wintertime polynya area and are persistent according to the present paper (Figure 2a). This underlines the possible role of East-Antarctic polynyas, e.g., the Lars-Christensen Coast polynya, for ice production and water mass modification.

[24] One application of the obtained polynya area is to calculate the associated ice and salt production. This could be done by combining it with the open-ocean net total heat flux derived from meteorological data taken, e.g., from the ERA40 Re-Analysis project of the European Centre for Medium-range Weather Forecast. The required open water

Table 2. Average Polynya Area for Sub-regions Ross Ice Shelf (RSP), McMurdo Sound (MMS), and Terra Nova Bay (TNB) and Its Standard Deviation for 1992–2002 as Given by *Martin et al.* [2007] and in the Present Study

Region	RSP	MMS	TNB
Martin et al. [10^3 km^2]	25.0 ± 5.0	2.0 ± 0.6	3.0 ± 0.8
This Study [10^3 km^2]	22.0 ± 4.5	2.0 ± 0.5	4.2 ± 0.8

fraction can be obtained either also with the PSSM using a second set of iterations that discriminate between thin ice and open water [*Markus and Burns*, 1995; *Kern et al.*, 2007], or by combining the polynya area with high-resolution ice-concentration estimates. It is suggested to thoroughly check model data to be used for this calculation against in-situ data, because ice-production values obtained for the Terra Nova Bay polynya using in-situ data on the one hand [*Van Woert*, 1999], and model data on the other hand [*Martin et al.*, 2007; *Tamura et al.*, 2008] do not agree properly.

[25] A second application of the obtained polynya area would be to use it for validating and/or initializing coarse-resolution ocean–sea-ice or sea-ice–atmosphere models to improve the representation of ocean-atmosphere interaction processes associated with polynyas.

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